

Predicting firm distress in high dimensions

A leakage-safe tabular benchmark: SIS, penalized regression,
SIR/SAVE, calibrated ensembles and promoted-model evidence.

3,728

FIRM-YEARS

422

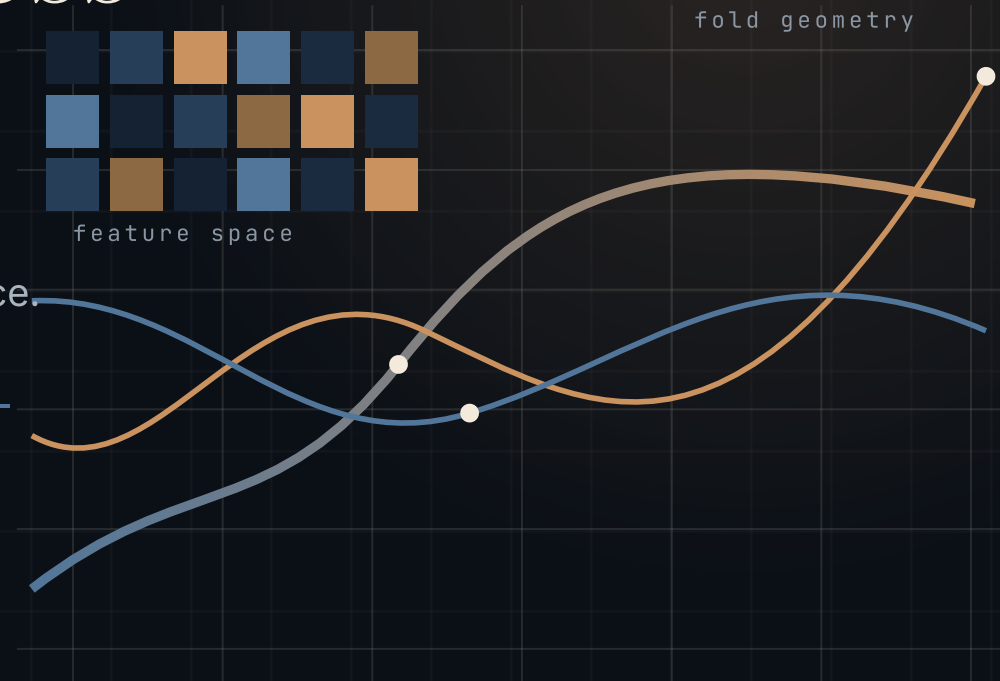
PREDICTORS

7

TARGETS

14

PROMOTED



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COURSE

Statistical Learning & Large Data

Prof. Francesca Chiaromonte · May 2026

THREE-LAYER PROGRAMME

L1 ·
TABULAR

L2 ·
FUNCTIONAL

L3 ·
ROBUST

Can a rigorous tabular model detect firm distress before it happens?

01

Which predictors matter?

SIS screening across 422 engineered features; stability selection across walk-forward folds.

02

Linear, dimensional reduction, or ensemble?

LASSO · Ridge · SIR/SAVE · 24 classifiers compared on the same holdout.

03

Does the signal survive 2 years out-of-time?

Hard temporal holdout 2022–2023 — never seen during model selection.

| Layer 1 scope: tabular scalar features only. Functional FDA methods (FPCA · FAS_tEN) are Layer 2.

What this presentation covers

DATA

01

Universe, collection & engineering

760 LSEG fields → 422 engineered predictors

SCREENING

02

Variable screening — 422 → ~90 stable

Stability selection across folds

METHODS

03

Penalized regression · SDR · 24 classifiers

Walk-forward CV · no firm-level leakage

RESULTS

04

Holdout AUC 0.83 · R^2 +0.16

All improvements statistically significant

EVIDENCE

05

Top-30 feature synthesis

Cross-method consensus: SHAP + permutation + screening

FUTURE

06

Layer 2: functional methods

FPCA + FAS_tEN on daily curves

European equities · STOXX 600 · 2016–2023


3,728
firm-years


483
firms

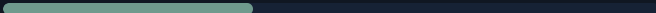

422
engineered predictors


7
prediction targets

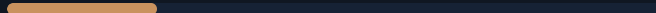
UNIVERSE

STOXX Europe 600
24 countries · 10 sectors

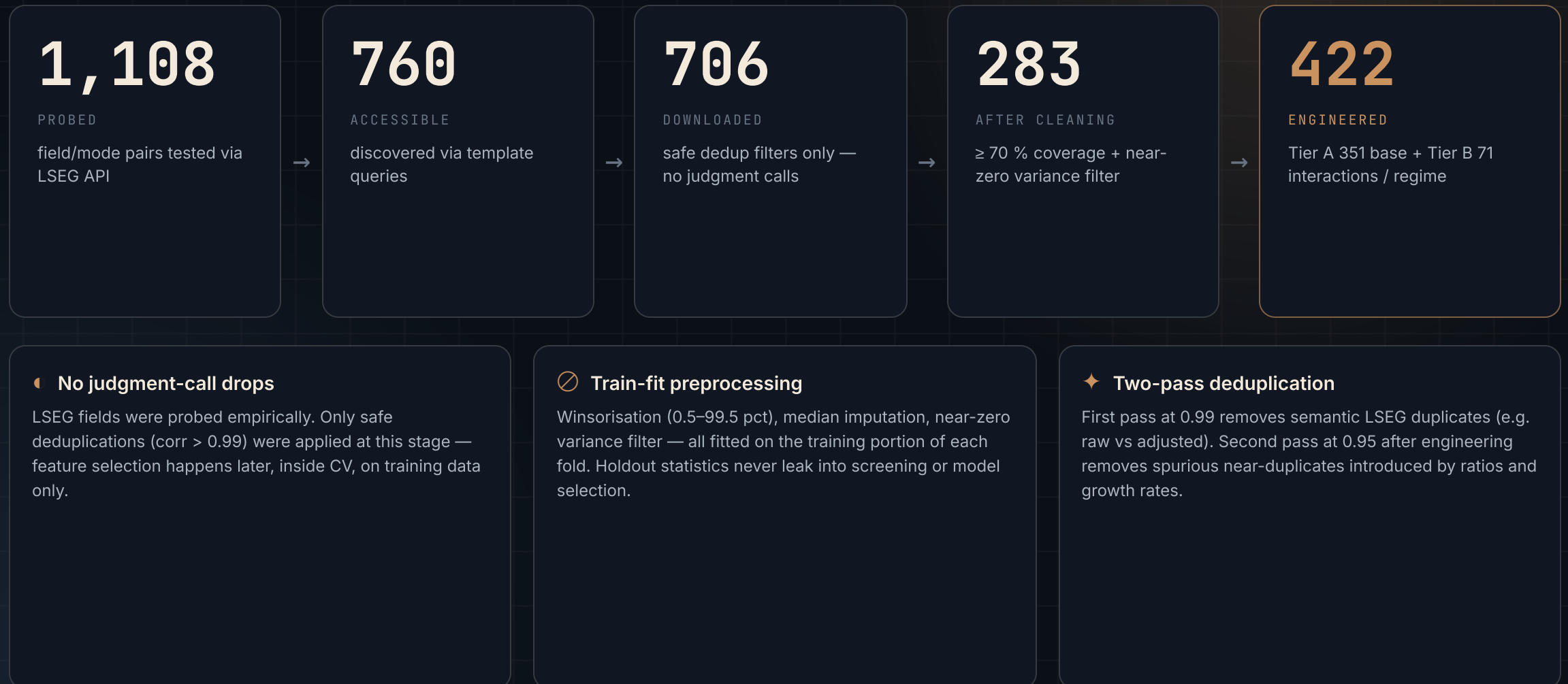
TRAINING PERIOD

2016 – 2021
2,765 firm-year observations

38% had drawdown >30% in this period

HOLDOUT (UNSEEN)

2022 – 2023
963 firm-year observations

23% had drawdown >30% — rate hike regime

From 1,108 probed fields to 422 predictors



422 engineered predictors across two tiers

TIER A — BASE PREDICTORS

351

Log-transformed originals

171`log(EBIT), log(Revenue), log(Total Assets)`

Asset-scaled ratios

121`log(X / Total Assets) for accounting variables`

YoY growth rates

22`Revenue growth, MCap growth, Dividend growth`

Curve-discretised stats

[→ Layer 2](#) **37**`ret_q25, vol_q05, max_drawdown, autocorr_1`

TIER B — ENGINEERED INTERACTIONS

71

Interaction terms

13`Beta × Volatility, MCap × Dividend yield, Debt × Liquidity`

Regime features

8`VSTOXX-quartile dummies, momentum-regime state, crisis flag`

Sector Z-scores

50`50 key variables standardised within each sector × year`

BRIDGE TO LAYER 2

37 curve-discretised statistics are scalar shadows of the full daily return / volatility / volume curves. Layer 2 replaces them with FPCA components and FASTen functional regression on the raw curves.

7 targets: absolute thresholds vs. cross-sectional ranks

REGRESSION TARGETS (CONTINUOUS BENCHMARK)

12-Month Return Peak-to-Trough Drawdown

Ridge $R^2 \approx +0.157$ on return (signal present); ≈ -0.008 on drawdown (path-dependent). Kept as control; classification answers the systemic-stress question.

High Drawdown $\geq 30\%$

absolute threshold

annual peak-to-trough $\geq 30\%$

prevalence $\rightarrow$ **7 - 83%**

Regime-dependent. 2019: Brexit + trade war. 2021: COVID recovery volatility.

Bottom 10% Returns

cross-sectional rank

worst decile of 12-month return within each year

prevalence $\rightarrow \approx$ **10%**

Always ~10% of the universe — stable across all regimes by construction.

Extreme Drawdown

cross-sectional rank

top decile of peak-to-trough drawdown within each year

prevalence $\rightarrow \approx$ **10%**

Analogous to Bottom 10% but on the drawdown axis.

PLUS AUXILIARY: Severe Distress $\geq 50\%$ (prev ~7 %) · Sector Laggards (within-sector rank)

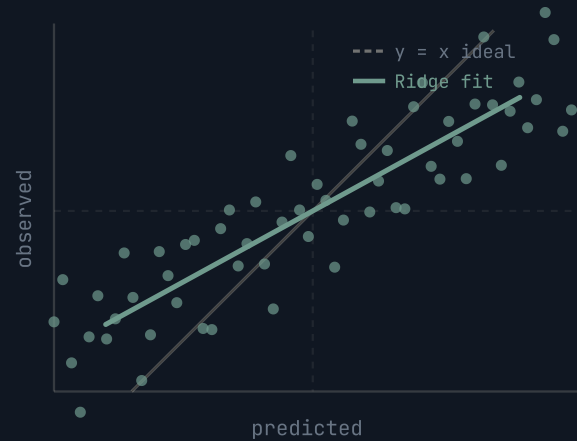
PREVALENCE BY YEAR — ABSOLUTE VS. CROSS-SECTIONAL



High Drawdown swings 7% $\rightarrow$ 83% across years — driven by market regime, not firm quality. Cross-sectional ranks test *relative* predictability and are always ~10%.

Continuous works — but answers the wrong question.

RETURN REGRESSION — HOLDOUT

Ridge R^2 +0.150

DM-significant signal on the continuous response.

DISTRESS ≥ 30 % — PREVALENCE BY YEAR

7 % → 83 %



Binary, regime-driven, sample size varies — exactly the signal classification handles.

CONTINUOUS

Continuous targets work — Ridge $R^2 \approx +0.150$ on return_12m, DM significant at 5 % FDR.

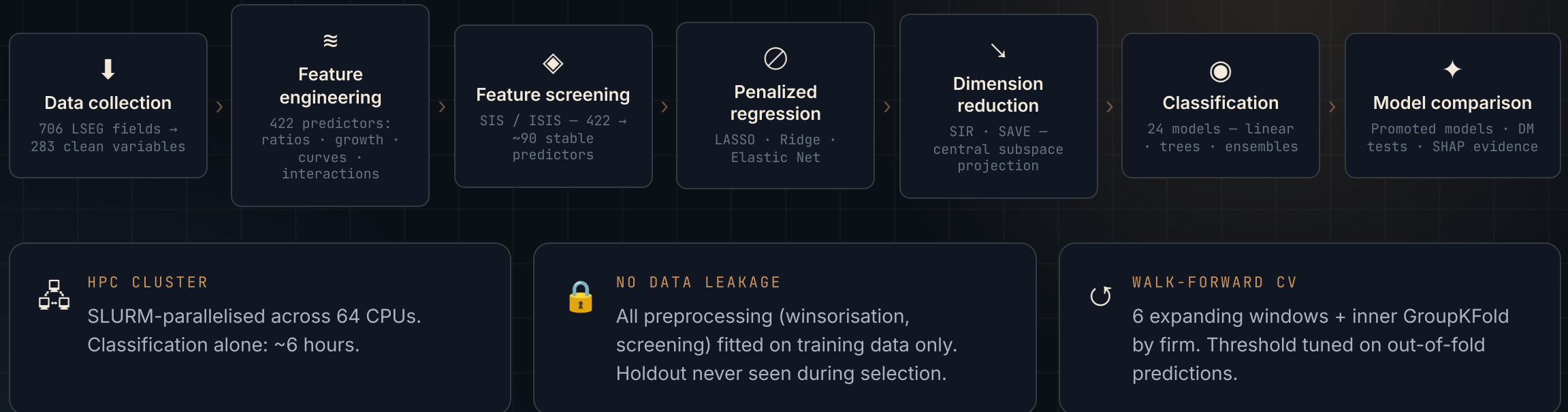
WRONG QUESTION

But *"how much"* $\neq$ *"did systemic stress occur."* The actionable signal for distress is **binary and rare** — regime-dependent across years.

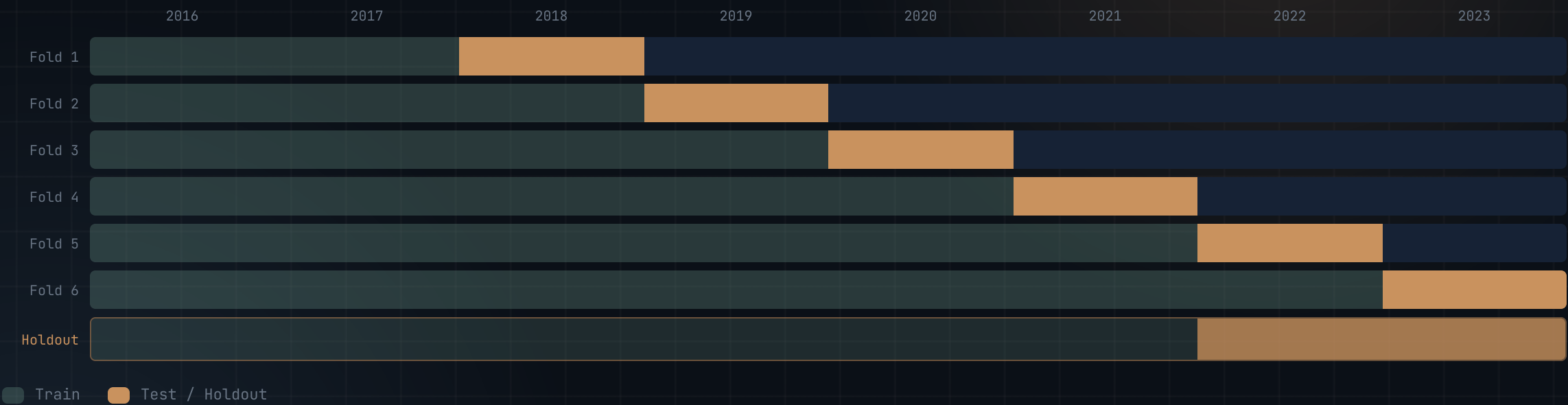
PIVOT

distress_30 becomes the **primary inferential target**. Regression retained as a *control* on the feature pipeline.

Seven-stage analytical pipeline



Expanding-window walk-forward + GroupKFold by firm



5-fold GroupKFold by firm · inner CV for tuning

Holdout 2022-2023 · never seen during selection

Drawdown $\geq 30\%$ rate fell 38% $\rightarrow$ 23% in holdout · rate-hike natural stress test

Four anti-leakage measures

⊘ Train-only deduplication

01

Correlation thresholds (0.99 and 0.95) computed on training observations only. Two passes: pre-engineering on raw fields, post-engineering on derived ratios.

WITHOUT IT → If computed on the full panel, future correlations would dictate which features survive — a subtle but real form of look-ahead leakage.

◆ Fold-wise SIS / ISIS screening

02

Sure Independence Screening (Fan & Lv 2008) runs inside each walk-forward window on the training portion only. Consensus sets are aggregated across folds.

WITHOUT IT → A single global screening would let test-period signal influence which 90 features make the cut — inflating downstream holdout AUC.

↘ Train-quantile winsorisation

03

0.5th–99.5th percentile thresholds fit on the train portion of each fold. Applied identically to test — no test quantiles ever computed.

WITHOUT IT → Using global quantiles lets extreme test observations modify the winsorisation cap, smoothing them artificially.

● Fit-on-train scaling + imputation

04

StandardScaler statistics, median imputation values, and range identification all fitted on training data only. Test set transformed with train statistics.

WITHOUT IT → Including test rows in scaling means the model sees the test distribution before prediction — a classic source of optimistic results.

| Deterministic transforms (log, asset-scaling, growth rates, curve discretization) are global — no cross-observation statistics, no leakage possible.

SIS / ISIS — 422 → 9 core variables

SIS · $D=2\sqrt{N}=105$
Marginal correlation ranking. 105 from 422 per target.

ISIS-LASSO
Joint iterative. 9 variables.

ISIS-SCAD
6 variables (subset of LASSO set).

STABILITY · 200 BOOTSTRAP
81 features at >90% selection frequency.

CONSENSUS SET
~95–102 per target. Used as screened_target feature set.



LASSO · Ridge · Elastic Net · Adaptive LASSO

RETURN_12M · R² HOLDOUT

METHOD	SET	R ²
Ridge	screened	+0.213 (walk-forward R ²)
Ridge	full	+0.157 holdout
LASSO	full	+0.152 holdout · 42 vars
Elastic Net	full	+0.149 holdout · 52 vars
Adaptive LASSO	full	+0.115 holdout · 21 vars
Mean Forecast	—	0.000 baseline

DM tests: Ridge + EN significantly beat mean forecast (BH-corrected).

MAX_DRAWDOWN_12M · R² HOLDOUT

METHOD	SET	R ²
SIR+OLS	full	-0.008 ← best regression
SAVE+OLS	full	-0.138
Adaptive LASSO	full	-1.010
Ridge	full	-1.380
Mean Forecast	—	-0.156 beats penalized!

max_drawdown is path-dependent and regime-driven. Regression is the wrong framing — classification works.

SIR / SAVE: finding the central subspace

SIR · LI 1991

Slice Y into H bins. Compute conditional mean $E[X|Y=h]$ per slice. The span of these slice means defines the central subspace — directions where X concentrates the regression signal.

SAVE · COOK & WEISBERG 1991

Uses conditional variance $\text{Var}[X|Y=h]$ instead of the mean. Captures directions where spread changes with Y . Complementary to SIR; tuned via inner GroupKFold (H slices, d dimensions).

HOLDOUT R^2

SIR + OLS -0.008 ← best regression

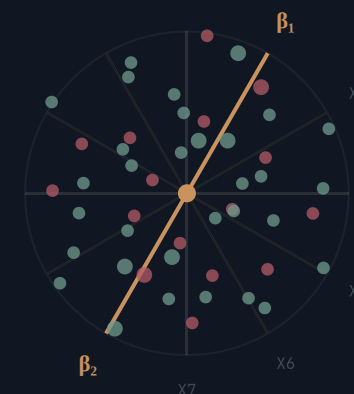
SAVE + OLS -0.138

Ridge +0.157 penalized benchmark

max_drawdown_12m: even the best SDR method can't recover signal lost to market noise.
Regression is a dead end for this target.

CENTRAL SUBSPACE · $X \in \mathbb{R}^{422} \rightarrow XB \in \mathbb{R}^2$

$X \in \mathbb{R}^{422}$ · predictor space



$XB \in \mathbb{R}^2$ · central subspace



24 model configurations across 3 tiers

TIER 3 · ENSEMBLES (3)

Voting Top-3

Voting Top-5

Stacking TopN

↑ fed by best base models

TIER 2 · WRAPPERS (6)

Calibrated LogL1 (Platt)

Calibrated LogL1 (Isotonic)

Calibrated RF (Isotonic)

Calibrated LGBM (Isotonic)

Calibrated XGB (Platt)

Balanced Bagging RF

↑ built on base models

TIER 1 · BASE MODELS (15)

Logistic L1

Logistic L2

Logistic EN

LDA

QDA

Random Forest

Extra Trees

HistGB

XGBoost

LightGBM

CatBoost

Gradient Boost

EBM

GaussianNB

KNN

■ FEATURE SETS

Three sweeps per target

Full (422) · Target-screened (~90) · Compact cross-target core (60)

● HYPERPARAMETER TUNING

Multi-objective Optuna

AUC + Brier Pareto front · inner GroupKFold by firm · 30 trials per (model × feature set)

◎ TARGETS

5 binary targets in parallel

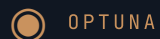
High Drawdown · Severe Distress · Bottom 10 % · Extreme Drawdown · Sector Laggards

⊘ THRESHOLD SELECTION

OOB cost-sensitive

41 thresholds evaluated · knee detection (Kneedle) on F1 · maximised on out-of-fold predictions to prevent holdout contamination

Optuna · Diebold–Mariano · Bootstrap + Conformal



OPTUNA

Multi-objective tuning

AUC + Brier · Pareto front

- Per-model hyperparameter search across both AUC and Brier objectives
- Inner GroupKFold by firm — no within-firm leakage during tuning
- OOF cost-sensitive threshold: 41 values evaluated, knee detection (Kneedle) picks optimal

24 models

5 targets

inner GroupKFold



DIEBOLD–MARIANO

Statistical significance

BH-FDR 5 % correction

- All promoted models tested against the prevalence benchmark
- Pairwise comparison between models on fold-wise squared loss / Brier
- Benjamini–Hochberg correction controls family-wise FDR at 5 %

DM test

BH FDR 5%

pairwise + vs benchmark



BOOTSTRAP + CONFORMAL

Uncertainty quantification

B = 1,000 resamples · MAPIE

- Bootstrap CI on holdout AUC, Brier, F1 — 1,000 resamples per metric
- Conformal prediction via MAPIE at $\alpha = 0.10$ and $\alpha = 0.20$
- Distribution-free coverage guarantees on calibrated probabilities

1,000 bootstrap

MAPIE conformal

 α 0.10 / 0.20

Annual return: real signal. Peak drawdown: intractable by regression.

● 12-Month Return

signal present ✓

MODEL	HOLDOUT R^2
LASSO	+0.152 42 vars
Ridge	+0.150 dense
Elastic Net	+0.149 52 vars
Adaptive LASSO	+0.115 21 vars
Mean forecast	-0.018 baseline

All penalized methods beat the mean forecast on the holdout; tuned via nested CV (walk-forward × GroupKFold by firm). DM significant at 5% FDR (Benjamini–Hochberg corrected).

● Peak-to-Trough Drawdown

regression fails ✗

MODEL	HOLDOUT R^2
SIR + OLS	-0.008 best available
Mean forecast	-0.156 naive baseline
Ridge	-1.380
LASSO	-1.010

All models lose to the mean forecast. Drawdown is path-dependent — it's driven by intra-year regime switches that annual snapshots cannot capture.

→ Pivot to classification: "did drawdown exceed 30%?" — AUC 0.83.

Holdout AUC 0.79 – 0.91 across five distress definitions

PRIMARY TARGET**High Drawdown**annual peak-to-trough $\geq 30\%$ **0.827**

AUC holdout

Lasso Logistic · all features

PR-AUC 0.612STABLE PREVALENCE $\approx 10\%$ **Bottom 10% Returns**worst decile within year
(cross-sectional)**0.877**

AUC holdout

Voting ensemble · feature-selected

PR-AUC 0.521STABLE PREVALENCE $\approx 10\%$ **Extreme Drawdown**top-decile drawdown within
year (cross-sectional)**0.873**

AUC holdout

Calibrated Lasso Logistic · all
features

PR-AUC 0.535LOW PREVALENCE $\sim 2.6\%$ **Severe Distress**annual peak-to-trough $\geq 50\%$ **0.908**

AUC holdout

Balanced Random Forest · feature-
selected

PR-AUC 0.220

60 FEATURES SUFFICIENT

Sector Laggardsworst decile within sector $\times$
year**0.786**

AUC holdout

Elastic Net logistic · parsimonious
(60 vars)

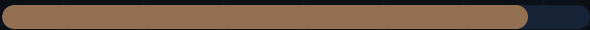
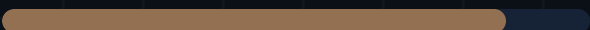
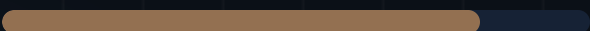
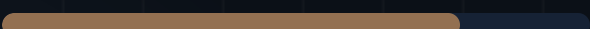
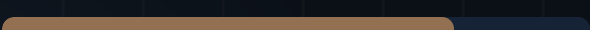
PR-AUC 0.339

Cross-sectional targets beat absolute thresholds — Bottom 10% (0.877) and Extreme Drawdown (0.873) both outperform High Drawdown (0.827). Stable $\sim 10\%$ prevalence every year.

Sparse linear structure dominates. Lasso Logistic or Elastic Net wins on 3 of 5 targets. Ensemble methods add value where linear falls short.

Top-30 features: three interpretable families

EVIDENCE SCORE · 4 METHODS × 14 MODELS (BREADTH × DEPTH)

Market cap growth		4.74	7/7
Beta (mkt sensitivity)		4.36	7/7
Market capitalisation		4.24	7/7
Dividend yield		4.11	5/7
Dividend yield growth		4.07	7/7
Market cap / Assets		3.85	6/7
1-year realised volatility		3.79	7/7
Equity-to-assets ratio		3.69	7/7
Cash-to-assets ratio		3.67	5/7
Return (25th percentile)		3.65	7/7

Sources: SHAP · Permutation importance · Screening consensus · SDR directions

MARKET / RISK

Beta · 1-year volatility · Market correlation · High/low range · Low-volatility episodes

Leading predictors — pure market signal, computed from daily prices

SIZE / MOMENTUM

Market cap growth · Market cap · Return percentiles · Historical drawdown

Cross-sectional positioning and trajectory over the year

FUNDAMENTALS

EBIT · Gross margin · Cash ratio · Debt ratio · Dividend yield

Incremental signal — not dominant alone, but stable across targets

Calibration, resolution, and threshold sensitivity

Brier Decomposition (Murphy 1973)

Uncertainty 0.234

irreducible base-rate term · higher = harder problem

Resolution 0.088

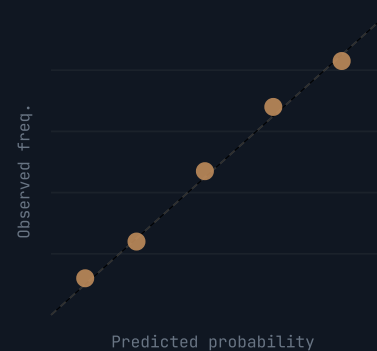
↑ better — model separates positives from negatives

Reliability 0.012

↓ better — miscalibration term: ≈ 0 means predicted prob. matches empirical rate

Brier = Uncertainty - Resolution + Reliability.
We want Resolution high and Reliability low:
the model both discriminates and predicts
probabilities that match observed frequencies.

Reliability Diagram · High Drawdown $\geq 30\%$



Well-calibrated after isotonic calibration.
Points near the diagonal.

Threshold Sensitivity

41 thresholds

Evaluated from 0.05 to 0.95 for every promoted model.

Optimal threshold ~ 0.32

Knee detection (Kneedle) selects balanced precision/recall trade-off.

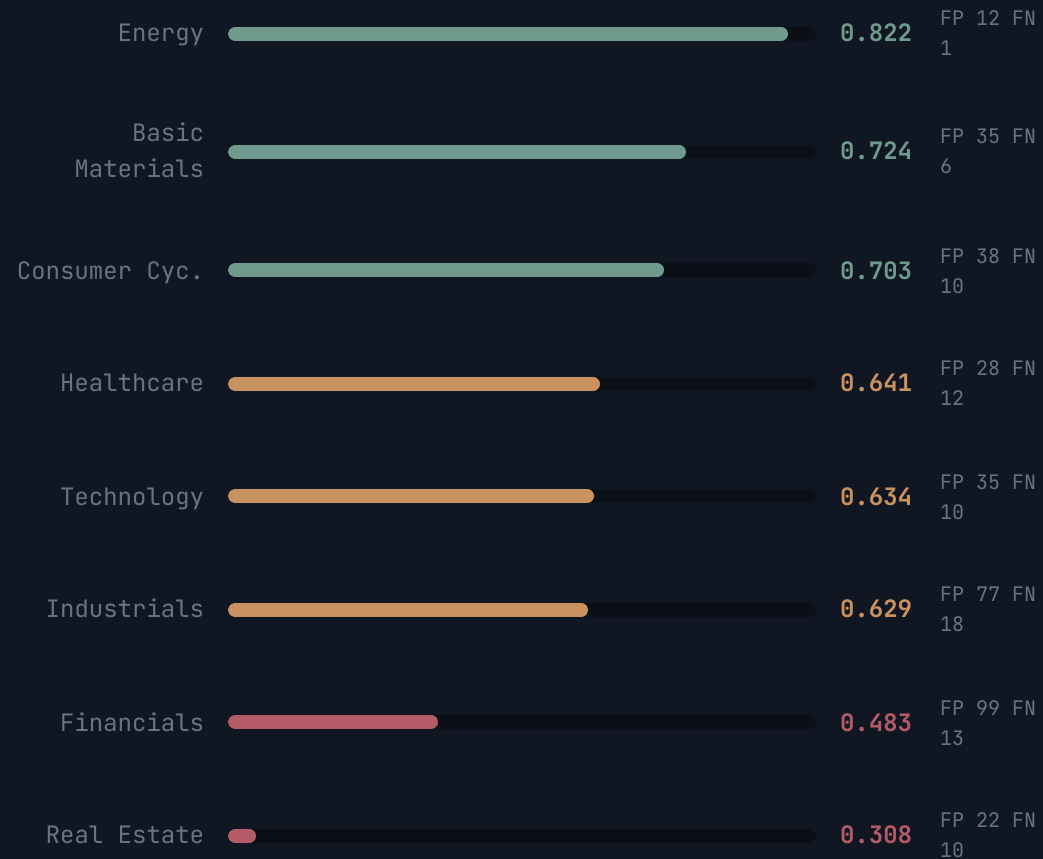
00F cost-sensitive

Final threshold maximizes F1 on out-of-fold predictions, preventing holdout contamination.

$F1 \approx 0.39$

At optimal: precision ~ 0.24 , recall ~ 0.85 . High recall is correct — missing distress is more costly than a false alarm.

What holds, what fails, and where

AUC BY SECTOR • LASSO LOGISTIC • HIGH DRAWDOWN $\geq 30\%$ • HOLDOUT

ECONOMIC READING

- ✓ **Energy (AUC 0.82)**
Commodity-cycle distress is well captured by market + fundamental signals.
- ✗ **Financials (AUC 0.48 • FPR $\approx 49\%$)**
Banks use leverage differently. Standard ratios over-signal distress. Sector-specific feature engineering needed.
- ✗ **Real Estate (AUC 0.31)**
Rate-hike sensitivity (2022–23) creates systematic misses that annual balance-sheet snapshots cannot capture.
- **Threshold drift across years**
Model calibrated pre-rate-hike over-predicts in 2023. Walk-forward recalibration per fold helps.

10-point methodological checklist

✓ **Temporal hold-out**

2022–2023 never seen during model selection or tuning

✓ **No firm-level leakage**

GroupKFold by firm in inner CV — same firm never in train and validation

✓ **Train-only preprocessing**

Dedup, winsorisation, screening + panel z (sector × year) — fitted on training only

✓ **Expanding-window CV**

6 walk-forward folds; full 422 + compact core 60 reported side-by-side

✓ **Diebold–Mariano tests**

Statistical significance vs. mean-forecast baseline · 5% FDR (Benjamini–Hochberg)

✓ **Bootstrap confidence bands**

1,000 resamples on holdout AUC, Brier score, and F1

✓ **Five prediction targets**

Primary + auxiliary targets — results consistent across all

✓ **Permutation test**

Shuffled labels → AUC $\approx$ 0.50. Signal is real, not spurious.

✓ **Probability calibration**

Platt scaling + isotonic calibration · reliability diagram shown

✓ **Cost-sensitive threshold**

Threshold optimised on out-of-fold predictions across 41 values

Layer 2: functional methods where scalar summaries fail

FPCA + FASTEN

FPCA PER CHANNEL

Extract functional principal components from return / volatility / volume curves (252 daily obs per firm-year). Replace 37 scalar curve stats with the full basis.

SCALAR-ON-FUNCTION REGRESSION

Regress targets directly onto FPCA scores. How many FPCs match or exceed Logistic L1 AUC 0.83?

FASTEN (FUNREGPOI • R) ZERO PAPERS IN FINANCE LITERATURE

Group-sparse functional regression. Selects which time windows within the year concentrate the distress signal.

Input ready: `curves_raw.parquet` • 3,728 × 758 • 252 pts × 3 channels • 0% missing

WHY FUNCTIONAL SHOULD WIN

Return percentiles, volatility episodes, and historical drawdown rank in the top-15 scalar features — but these are 1D projections of the full curve. FPCA extracts the complete basis.

Annual peak drawdown is unpredictable by regression ($R^2 \approx 0$). The intra-year shape — crash-and-recover vs. slow decline — may hold the signal that annual snapshots miss.

FASTEN will identify which calendar window (Q1? Q4?) concentrates the distress signal — invisible from year-end balance-sheet statistics.

Where tabular fails → functional should win.

Layer 1 at a glance

DATASET

3,728

firm-years · 483 firms · 2016–2023

422

engineered predictors (Tier A 351 + Tier B 71)

STOXX 600

24 countries · 10 sectors · European equities

7

prediction targets evaluated

BEST HOLDOUT RESULTS

12-Month Return

regression · control

R^2 **+0.150**

High Drawdown $\geq 30\%$

classification

AUC 0.827

Severe Distress $\geq 50\%$

classification

AUC 0.908

Bottom 10% Returns

classification

AUC 0.877

Extreme Drawdown

classification

AUC 0.873

Peak-to-Trough Drawdown

regression

$R^2 \approx 0$

TOP PREDICTORS (ALL 7 TARGETS)

1. Market cap growth · 7 / 7 targets
2. Beta (market sensitivity) · 7 / 7
3. Market capitalisation · 7 / 7
4. 1-year realised volatility · 7 / 7
5. Dividend yield growth · 7 / 7

... 25 more in top-30 consensus set

SIGNAL HIERARCHY

Market / Risk > Size / Momentum > Fundamentals

Classification is the primary track for systemic-stress targets (AUC 0.83–0.91). Regression on continuous return retained as a control on the feature pipeline (Ridge $\approx +0.150$).

Layer 2 next: FPCA + FAS_TEN — functional methods on daily return / volatility / volume curves for the hard targets.

References

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PENALISED REGRESSION

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CV & INFERENCE (ADAPTATION)

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FEATURE SCREENING

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- ◆ Saldana & Feng (2018) — SIS R package · JSS
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SUFFICIENT DIMENSION REDUCTION

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 - Platt (1999) — Sigmoid calibration · Adv. in Large Margin Classifiers
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Thank you · Questions?

Questions?

COURSE

Statistical Learning & Large Data

Prof. Francesca Chiaromonte · SSSA + UniPi · May 2026

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NEXT

Layer 2: FPCA + FAS_tEN on daily curves

Functional methods on return / volatility / volume curves (252 days × firm-year).

Layer 3: robust functional SDR for distribution-shift resilience.

STACK & REPRODUCIBILITY

DATA LSEG Workspace · 706 fields · 252-day price curves

CODE Python 3.12 · scikit-learn · LightGBM · XGBoost · CatBoost · R (SIS, dr)

HPC SSSA cluster · SLURM · 64-CPU classification stack run

ENV pyproject.toml + uv lockfile · random_state=42 throughout